Chapter 5

Automated Identification of Lectin Fine Specificities from Glycan-Array Data

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Lectins are proteins that mediate biological functions through the recognition and binding of specific oligosaccharides. Lectins also are widely used as probes to detect glycan ligands in biological samples. The specificities of lectins (i.e., the patterns of glycans bound by each lectin) are highly complex. Glycan-array technology has been transformative for the study of lectin-glycan interactions, together with the increasing range of glycans available through new synthetic methods, but the significant difficulties in analysis have hindered the full use of the data. This chapter provides detailed information about a new analysis tool, the MotifFinder analysis platform, to meet this need. The MotifFinder analysis platform is based on several innovations. It uses a novel framework for representing lectin specificity as a family of motifs organized as primary motifs and fine specificities. Each motif is defined using the new syntax that permits precise specifications of the complexity and tolerance of a pattern, and it employs regression trees with simultaneous motif optimization to find the family of motifs that accurately describe lectin specificity. Furthermore, MotifFinder can jointly analyze multiple datasets to increase the completeness and accuracy of the output. The output formats are usable as standalone results or as inputs for subsequent analyses. This tool could aid researchers in enhancing their understanding of lectin specificity and fully capitalizing on the increasing range of glycan-array data.

Introduction

Lectins are proteins that bind to carbohydrates. They do not induce a structural change to the glycans, like glycosidases and glycosyltransferases, but rather they serve to recognize and bind glycan features that fit the binding pocket of the lectin. This function is used in various ways, such as immune-cell recognition, organization of epithelial membranes, and regulation of cell migration. The protein-glycan interactions mediated by lectins are thus at the core of biology. For researchers studying such interactions, determining the binding preference (or the specificity of a lectin) is fundamentally important. This task is surprisingly difficult though; relative to other classes of

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biomolecules such as nucleic acids and proteins, comparatively little is known about the details of the glycan structures that mediate the interactions.

Each lectin has its own preference for patterns of glycans. The rules governing these preferences can be obscure since the contact between lectin and glycan may tolerate variety in structures or involve monosaccharides that are noncontiguous or on different branches. As a result, empirical testing of lectins binding to many different oligosaccharides is needed to obtain reliable and detailed information about lectin specificity. Glycan-array technology has greatly helped with this task because it enables researchers to measure the binding of a lectin to many different glycans in parallel. This technology is further bolstered by ongoing developments in glycan synthesis methods that are producing glycans with increasing complexity. Several groups and companies are now generating microarrays containing hundreds of discrete glycans spanning multiple biological contexts, such as microbial glycans (*1*–*3*), sialic acid–containing glycans (*4*–*6*), mammalian glycans (*7*–*11*), and other glycans (*12*, *13*). The glycan arrays from the Consortium for Functional Glycomics (CFG) (*14*) were especially valuable and have been used in several thousand experiments. Several reviews provide background on various types of glycan arrays and experimental platforms (*15*–*20*). These datasets present great opportunities in the study of lectin-glycan interactions, and they also present new obstacles in the analysis of the data.

The Need for Automated Analyses of Glycan-Array Data

The manual analysis of glycan-array data is inadequate for fully utilizing the available datasets. Manually identifying the features that influence binding positively or negatively would require a painstaking, lengthy review by an expert, and even expert analysis is subject to inaccuracy, incompleteness, or bias. In addition, the analysis would be inaccessible to non-experts, and the processing time would not facilitate routine analyses. Automated analysis methods could improve accuracy and throughput, while enabling the integration of information from multiple experiments. This latter point is important because no single array contains all the glycans that could be relevant for a lectin. Each array has limited size and is often specialized to a family of glycans. In addition, the joint analysis of multiple experiments is needed to account for variability and concentration-dependent effects. The specificity of a lectin may shift between concentrations due to cooperative, multivalent effects, which could potentially be assessed through parallel analyses of experiments performed using a range of lectin concentration.

This chapter provides detailed information about a new platform for the automated analysis of glycan-array data. This method uses a new syntax for representing glycan motifs (*21*) and a new approach for representing the overall specificity of a lectin. The software is available for download as the MotifFinder analysis platform (*22*). Notable features of the system include: automation in the analysis of glycan arrays; automated generation of precisely defined motifs; the ability to integrate information from multiple experiments; and output formats that are usable as standalone results or as inputs for subsequent analyses. With the download of MotifFinder, users are provided with a detailed user's manual that provides the technical details of the algorithms in greater detail than can be described in this chapter. Additionally, users are provided with step-by-step tutorials with screenshots on the use of MotifFinder and all the features highlighted in this chapter.

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Fundamentals of the Motif Finder Analysis Platform

Using New Methods for Representing Motifs and Lectin Specificities

The platform is built on a new framework for representing the specificities of lectins. The framework accounts for the fact that a lectin does not bind a single glycan substructure but rather a family of motifs or patterns of substructures. Within the family, some motifs are closely related, and others are dissimilar; some are bound strongly and others weakly. Therefore, to present the family of motifs comprising a lectin's specificity, we group the motifs into two types: primary motifs and fine specificities (Figure 1). The primary motifs are distinct from one another and could be found on nonoverlapping groups of glycans. For example, the lectin concanavalin A binds mannose in N-linked glycans, and alternatively it binds polysaccharides of glucose in an alpha linkage. The fine specificities are subsets of the primary motifs. They represent narrower definitions of each primary motif, in which the fine specificities have differences between them in their level of lectin binding. For example, using glycan arrays containing human milk oligosaccharides, such as the *Aurelia aurantia* lectin, bound terminal, alpha-linked fucose and bound particularly well when two instances of this motif were present on different branches from one another (*21*). The latter definition is a narrower subset of the former, and so it represents a fine specificity of *A. aurantia* lectin.

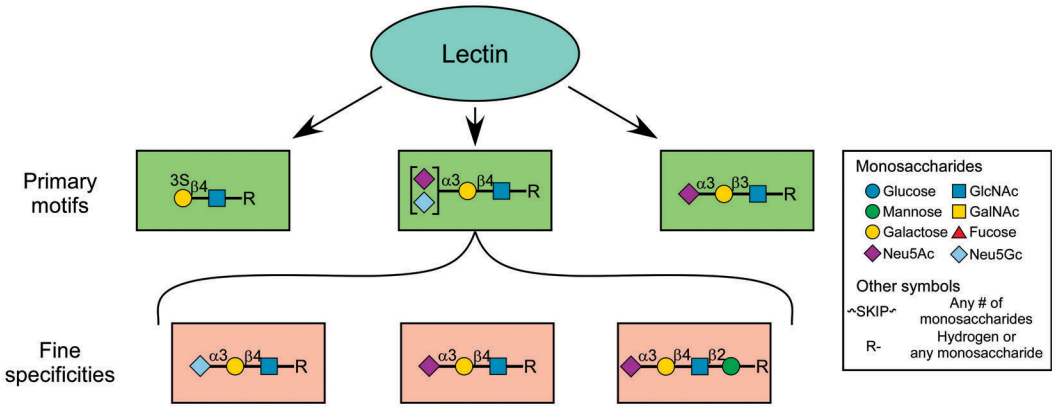


Figure 1. Representing lectin specificity as families of motifs. The primary motifs are the main specificities, and the fine specificities define gradations in binding within a primary motif.

Another feature of our representation of lectin specificity concerns the syntax for the individual motifs. A motif is not equivalent to a fixed substructure; it is a pattern. As a pattern, it encompasses variability. For example, some lectins that bind sialic acid can accommodate any of the many versions of sialic acid, while others bind only specific versions in precise linkages. The specificities of both types include sialic acid, but the differences in tolerance must be defined. To permit such definitions, we utilized a new syntax (*21*) for glycan motifs (Figure 2A). The syntax uses embedded wildcards, logical operators, and special symbols to enable the definitions of almost any type of variability or complexity. This improved motif syntax removes ambiguity through explicit definitions of tolerance for variations in monosaccharide identity, linkage carbon, and monosaccharide anomeric state (Figure 2A). In addition, we used the novel feature of "skips" to allow two or more motifs to be linked by a variable number of undefined monosaccharides. This powerful feature provides for motifs that describe effects from different branches or different core structures.

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The motif family of a lectin can be graphically represented as a tree, in which the main branches indicate the primary motifs and the subsequent splits indicate the fine specificities (Figure 2B). For the visualization of glycan-array data, the glycans can be grouped according to their possession of each motif, which enables comparisons between the amounts of lectin binding to each group of glycans (Figure 2C). The fine specificities represent gradations in binding within the range of the parent motif.

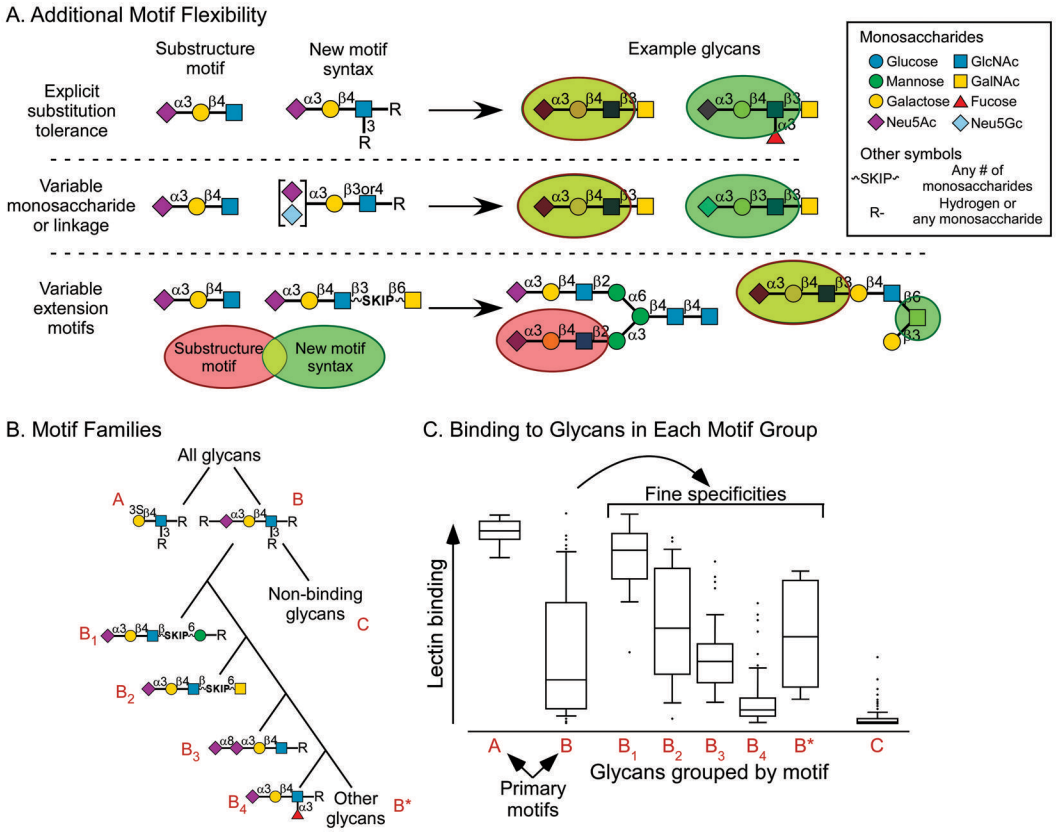


Figure 2. Motif flexibility and organization. (A) The new motif syntax enables precise definitions of variability and tolerance using wildcards and terms such as "Skip.". The glycans containing particular motifs (right) can be more precisely defined using the new syntax than with simple substructures. (B) Tree structure of a motif family representing the specificity of a lectin. (C) Relationship between lectin binding and motifs. In the simulated data, the glycans in a glycan-array dataset are grouped by the motifs they contain, and the amount of lectin binding to the glycans is plotted. Lectin binding is different between the primary motifs A and B, and the broad range of binding to motif B can be explained by the fine specificities, B_1-B_4.

The system described above is useful for describing lectin specificity, but the significant challenge in automated analysis was developing an algorithm for identifying accurate motifs and tree structures. We met this need through a model-first approach. Unlike methods such as frequent subtree mining, which generate motifs and then assess their value in a model (*23*, *24*), the model-first approach begins by attempting to create a tree structure (or model) for lectin binding and then developing motifs guided by their value for the model. This approach obviates the definition of all viable motifs, which would be impractical given the huge number of options available through the motif syntax. A consequence of this approach is that the resulting model is unlikely to be the optimal

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model, which cannot possibly be calculated, but will be comparably accurate with a short calculation time. We selected the regression tree method to execute the model-building process based on its compatibility with the representation of lectin specificity as a family of motifs.

Identifying Families of Motifs to Describe Lectin Specificity: Regression Tree Models

The regression tree is a machine-learning algorithm that is simple yet remarkably powerful. The method iteratively splits the observed data (the amount of binding to each glycan) using the available features (the motifs) based on the associations of the features with variations in the data. In this case, the glycans on an array are initially divided based on the presence or absence of the motif that produces the greatest difference between the groups in lectin binding (Figure 3). The separate groups of glycans are searched for motifs that further associate with binding strength within each group, either as alternative primary motifs or as fine specificities. The overall ability of the regression tree to explain the variations in the data is assessed by a fit parameter of R^2. The R^2 can be calculated as the square of the correlation, r, between fitted and observed binding, where fitted binding is the mean binding of glycans with their respective motifs. The process of splitting the data continues until any additional splits give an unacceptably small improvement of R^2. The result of this process is a tree model for the binding of the lectin. The regression tree algorithm is uniquely suited to our needs as it inherently identifies both primary motifs and fine specificities.

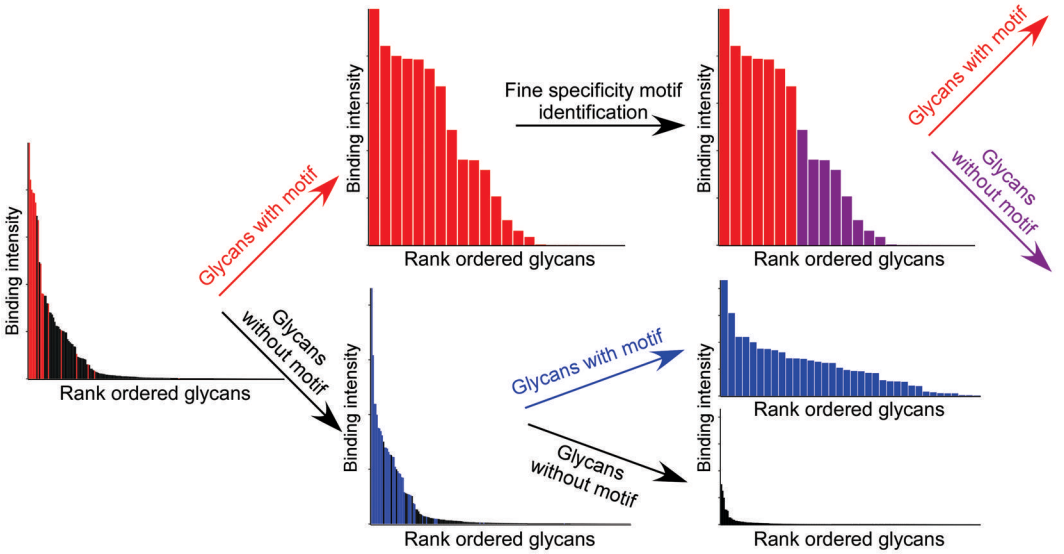


Figure 3. Regression trees to split the observed data by the presence and absence of a motif. The first split finds the top primary motif, and further splits are evaluated only in the context of the new datasets. Splitting the sets with the primary motif allows for the identification of fine specificities and splitting the sets without a motif allows for the identification of an alternative primary motif.

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Integrated Optimization of Motifs in the Model-Building Process

The algorithm begins with a large stock of motifs that are predefined and also generated from substructures of the glycans on the array, but modifications to the motif definitions are necessary to account for nuances in lectin specificity. We integrate the process of motif optimization with the model-building process in order to develop the most valuable motifs and model (Figure 4A). Through model-driven motif optimization, it is possible to learn the correct motif from the data rather than relying on the best motif having already been defined.

For primary motifs, the top-fitting motif from the initial list is used as a lead (Figure 4B) while for fine specificities, the primary motif is used as the lead. Multiple types of optimization are performed. At each type of optimization, the optimization continues until any changes result in a reduced score. The first step is to test interactions between the top motif and other stock motifs. We do this through the "skip" notation. Two motifs can be combined in a variety of ways using skips, such as placing them on separate branches or in a linear arrangement. The next step is testing the growth of the motif by a stepwise addition of monosaccharides. At first, monosaccharides are added using a randomly selected glycan containing the top motif to guide the monosaccharide addition. The next step is testing whether restricting the occupation of specific sites improves the score. This is done through the <f> notation, which indicates that a location is free (i.e., unoccupied by any monosaccharide or sulfate or phosphate group). Finally, more monosaccharides are added where each monosaccharide is added nonspecifically (any monosaccharide, any linkage carbon, and any anomeric arrangement). This is done without the use of a guide glycan, limiting the influence that the random selection of a guide glycan has on the optimization. The nonspecific additions are made as specific as possible based on the glycans on the array in order to make the motif as specific as possible without reducing the fit. This nonspecific addition resolved by array glycans allows us to test a variety of modifications, including the locations of the linkages (for example, Galβ1-4GlcNAc vs. Galβ1-3GlcNAc), the stereochemistry of the linkages (alpha vs. beta), and variation in the monosaccharides (for example, galactose vs. glucose). The completion of nonspecific additions is followed by a final round of restricted substitution tolerance testing in the same manner as was done for other specific additions. This multistep motif optimization allows the program to precisely define the motif, at once favoring simple motifs while also achieving the level of flexibility needed to accurately describe the tolerated variations.

A concern with model generation is "overfitting" to features that are present in training datasets but are not general to future datasets. The MotifFinder analysis platform provides the option to adjust the stringency of the model through two parameters. One is the complexity parameter, which defines the minimum change in R^2 required for a split; the other is the split size, which defines the minimum number of glycans in each branch of a split. The defaults use high-confidence settings, but expert users may wish to use looser settings to explore subtleties. For example, based on the knowledge that a particularly important motif occurs on only two glycans on the array, a user may want to examine minimum splits of two. Another method to reduce overfitting is cross validation. The software incorporates this method to remove splits that do not meet a minimum cross validation error. Further information about these steps and other advanced features are provided in the release notes.

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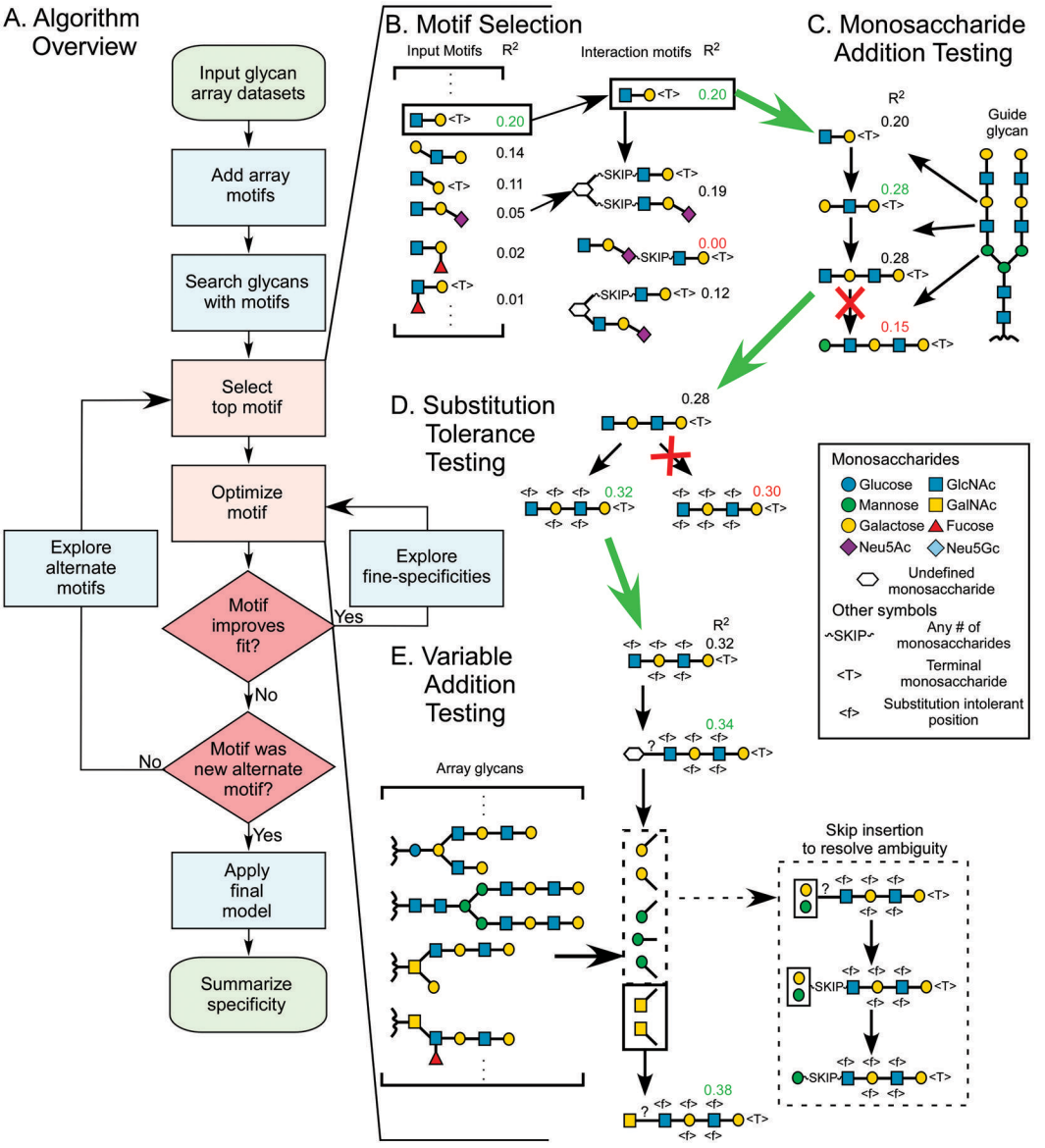


Figure 4. Model generation with motif optimization. (A) Beginning with one or more glycan-array datasets, the MotifFinderautomated analysis algorithm builds up families of motifs through optimization of stock input motifs. (B) The best fitting (R^2) motif is chosen from a stock list of motifs. Possible interaction motifs are tested by evaluating their combined fit and a number of potential structural relationships (linear or branched). (C) The selected motif is optimized by the addition of new monosaccharides, as they are found on the guide glycan. (D) Each potential substitution site on the motif is tested for substitution tolerance and defined as substitution intolerant (<f>) if doing so does not reduce the fit. (E) Positions which can tolerate substitutions are tested for variable additions through the addition of undefined monosaccharides with undefined linkages. All array glycans with the new motif are evaluated for the monosaccharide and linkage options possible for the unspecified components of the motif, and the best (in terms of fit) subset is chosen and added to the motif. In cases where excessive ambiguity is retained, insertion of a skip is tested for the potential to resolve some ambiguity.

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Integrating Information from Multiple Experiments

In the development of a model, the more data available to learn from, the better the model will do in uncovering the actual reality underlying the observed data. The combined analysis of multiple glycan-array datasets is difficult due to the differences between the experiments in conditions and methods. We addressed this challenge by using a multivariate approach in which each of the dependent variables (each glycan-array dataset) is analyzed simultaneously. Instead of selecting motifs that best fit a single array, motifs are selected for their cumulative fit to all arrays. The method uses a maximum likelihood estimator to find rescaling and shifting parameters to map one array onto the other (Figure 5).

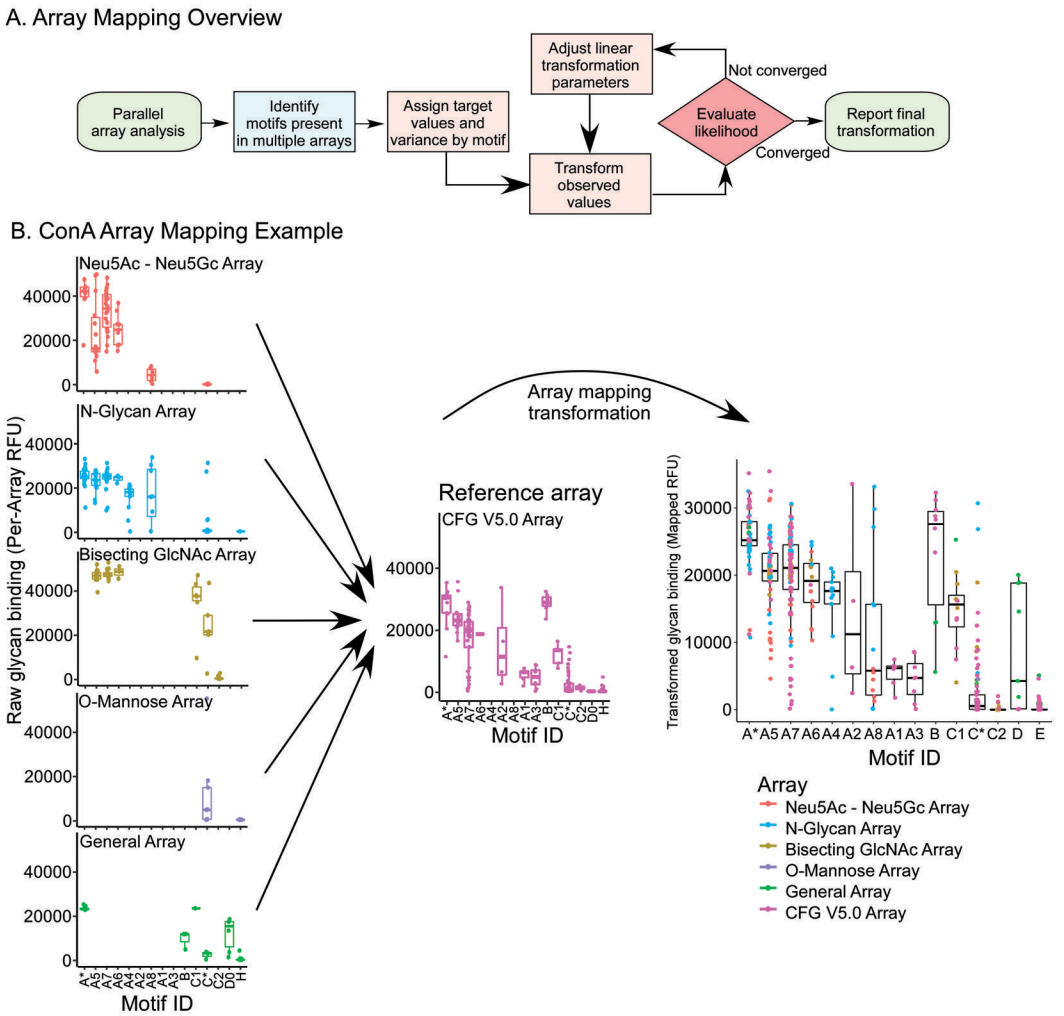


Figure 5. Joint analysis of multiple arrays. (A) Mapping arrays. Glycan arrays from different platforms and of the same lectin concentration can be mapped onto a reference array through maximum likelihood estimation of linear transformation parameters. The optimal linear transformation parameters are chosen with respect to the likelihood of observing transformed values given normal distributions with the motif value and standard deviation calculated from the glycans on the target array with the motif. (B) A number of arrays can be mapped onto the scale of a single target array. This allows for comparison of array-specific motifs (motifs A4 and A8) to more general motifs.

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Our implementation is comparable to the XPN method for cross platform normalization of gene expression microarrays (*25*) with a number of accommodations made for the greater differences between glycan arrays both in platform and glycan content. We compare array groups of glycans that contain the same motifs. "Likelihood" is the probability that the observed values in the comparison array could arise from normal distributions derived from the first array. An optimization algorithm is used to solve for rescale and shift parameters that are applied to the comparison array to give the maximum likelihood of obtaining the observed data. The method does not force all motifs to be similar in relative binding between arrays because in some cases, the binding could be legitimately different between arrays based on the glycans or conditions. To prevent a motif that is weakly bound in either array from having undue impact, a weighting factor is applied to the rescale parameters based on the product of the binding levels of the glycans containing a given motif.

This approach is based on several assumptions. It is assumed that the differences between arrays can be corrected with a linear transformation and that binding to the motifs is not affected by differences between the arrays in the types of glycans. These assumptions could be wrong in some cases. For example, a lectin that recognizes terminal N-acetyl-lactosamine could display different binding to an array containing N-linked glycans than to an array containing human milk oligosaccharides. Therefore, analyses linking arrays with divergent content should be used qualitatively unless caution is taken to make sure the assumptions hold true.

The potential value of this approach was shown in an analysis of the lectin concanavalin A (Figure 5B). Datasets were available from six different glycan arrays. The glycans were minimally overlapping between the arrays, and the raw values quantifying lectin binding had diverse ranges. Five of the datasets were mapped onto the reference dataset (the one from the CFG version 5.0 array) to produce an integrated output of the relationships between lectin binding and the presence of various motifs. Some of the top motifs were represented across datasets while others were in only one or two datasets, which show that the integration provided a broader analysis than would be possible using only one dataset with greater support for relative values between motifs.

In addition to the limitations inherent to working with data from different datasets, users must be aware of experimental differences between arrays. For lectins, in particular, it is important to distinguish between providers as there are frequent differences in affinity and specificity between the lectins of different providers. Differences in experimental conditions can also produce systematic effects on specificity and affinity. Finally, while the analysis of multiple concentrations on the same array is valuable and recommended, the display and discussion of results often must focus on a single concentration to give a detailed exploration.

Motif Finder Automated Analysis Compared to Historical Analysis Tools

A number of approaches for the analysis of glycan arrays have been developed. Several have used graph theory approaches, such as subtree mining (*23*, *26*) and subtree fingerprints (*24*, *27*–*30*), to identify subtrees common among bound glycans but not among nonbinding glycans (where binders and nonbinders are determined through a threshold). Other tools extend the motif identification methods from proteins and DNA to tree structures. These include the Hidden Markov Models (*31*), multiple sequence alignment (*32*, *33*), and q-grams (*34*). MotifFinder is unique in using model building to drive motif mining.

Several features of the MotifFinder automated analysis platform are advantageous. Because the method does not dichotomize glycans into binders and non-binders, but rather uses the continuous variations in glycan binding, it allows identification of fine specificity gradations in binding. The only

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other approach that does not dichotomize binding is the partial least squares regression method introduced by Xuan et al. (*27*) MotifFinder is the only software that provides integrated analyses of multiple datasets and the ability to capitalize on data from multiple lectin concentrations. Cholleti et al. demonstrated the use of multiple lectin concentrations by adding up the signals for each glycan for all concentrations (*23*), but this method limits the added information and biases the results in favor of the arrays with the highest signals. Finally, the motif syntax employed by MotifFinder enables the representation of nearly any unusual, unforeseen characteristic of lectin specificity. A few other methods incorporate some complexity in motif definitions. For example, Coff et al. introduced a subtree mining method featuring "restricted linkages," which are substitution-intolerant monosaccharide carbons (*24*); several sequence alignment methods (*31*, *32*) offer variable monosaccharides and linkages with the ProfilePSTMM method, in particular, allowing for long distance interactions (through insertions). In contrast, the MotifFinder syntax offers nearly unlimited flexibility and detail. Few of the previously cited comparison algorithms have been packaged into accessible software programs. Most important for the broader research community, MotifFinder is available for free download, has an intuitive interface and output, and handles any type of glycan-array data.

Output and Application

Example Output: RCA-I

The *Ricinus communis* agglutinin 1 (RCA-I) from the castor bean plant is a strong binder to galactose. The basic understanding of its specificity dictates that it primarily binds terminal galactose in a beta-4 linkage, although the advent of glycan-array datasets showed that its specificity is somewhat more complicated. The datasets provided evidence that RCA-I can bind galactose in some cases when the 6′ carbon or other carbons are occupied or potentially when the galactose is alpha-linked but assigning clear trends has been elusive.

While numerous glycan-array experiments have been performed using RCA-I for the sake of this example, MotifFinder was used to analyze only the vector lab–supplied RCA-I analyzed by Dr. Lara Mahal on the CFG array version 5.0, including all the available concentrations (0.1, 1, 10, and 100 μg/mL). MotifFinder found 7 primary motifs, 4 of which also had fine specificity motifs (13, 3, 3, and 2 fine specificity motifs), for a total of 24 motifs in the regression tree (Figure 6A). The top primary motif was as expected: a terminal galactose in a beta-4 linkage with substitution tolerance on the 6-carbon of galactose. The others show some information about alternate specificities. Motifs B and C have a sulfate at the 6′ of Gal or GalNAc, and Motif D is similar to the top primary motif except the GlcNAc that links directly to the linker substrate. Motif E is somewhat surprising because it shows the galactose with the 3′ or 4′ carbon occupied, in this case, by alpha-galactose. This finding suggests that RCA-I either tolerates 3′ or 4′ substitution of the Gal if the substitution is alpha-linked galactose or that RCA-I can bind alpha galactose if it is linked to a 3′ or 4′ carbon in the absence of the fucose found in blood group antigen B. Motif F is also surprising because it shows Gal in a beta-3 linkage, but not in a beta-4 linkage. It also shows the Gal on two separate branches, suggesting that this weaker motif is aided by multimeric binding. Motif G shows 6′ substitution of the Gal with a sialic acid without the favored presentation of the unsubstituted LacNAc disaccharide.

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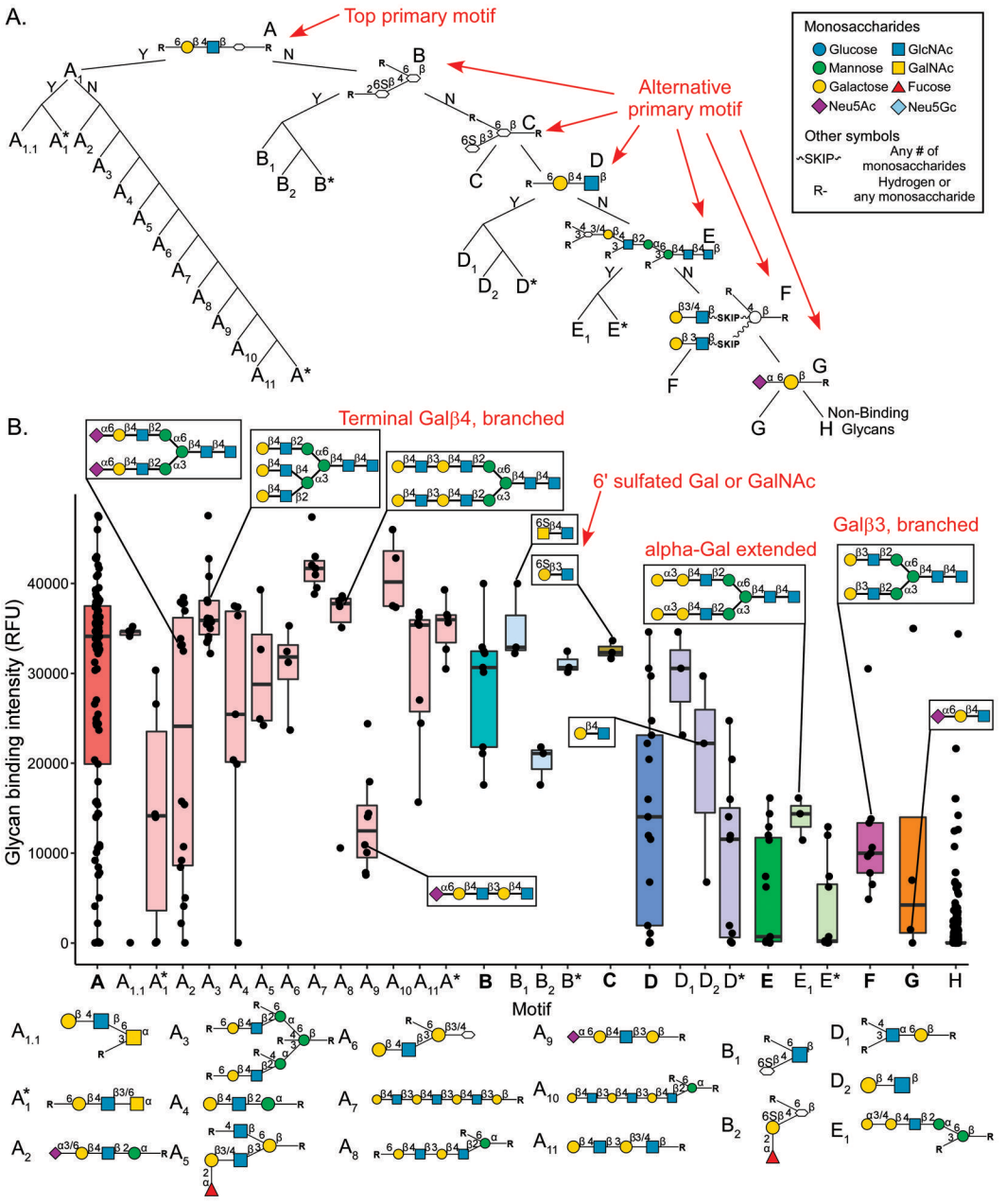


Figure 6. Example output. (A) The complete family tree produced through the analysis of CFG array data (all concentrations) for RCA-I from Vector Laboratories and analyzed by the laboratory of Dr. Lara Mahal. Motif structures shown here were generated using the DrawGlycan Tool (35). (B) The observed binding for RCA-I measured at 10 μg/mL (CFG ID: "primscreen_4666"), grouped by motif. Example glycans for various motifs are highlighted.

The associations of the motifs with binding can be examined by grouping the glycans according to motif and plotting the levels of lectin binding to the glycans (Figure 6B). The glycans with the primary motif A span a huge range of binding, but the fine specificities sort out some of the distinctions. For example, the glycans with fine specificity motif A_8 have higher binding than those

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with fine specificity motif A_9. A review of the motifs and the example glycans lead to the conclusion that RCA-I binding to Galb1-4GlcNAc is strongest when the motif is present on two or more branches (with a preference for poly-N-acetyl-lactosamine), and it is weaker when the Gal is 6′ sialylated. The distinctions in binding between the remaining primary and fine specificity motifs support the idea that the motifs indicate the features that modulate RCA-I binding. Notable outliers exist in the nonbinding group; these outliers are glycans that have conflicting results or results for which not enough support exists to form an additional motif. The greatest outlier is a ganglioside "GQ1b," which is inconsistent as RCA-I shows no other binding to gangliosides. The second highest outlier is a Type 2 LacNAc with a phosphate group on the 6-carbon of the GlcNAc, which could be a true binder that does not get assigned a motif due to underrepresentation on the array. Beyond these, a number of lower binding glycans exist in the nonbinding group including various blood epitopes and Lewis-terminated LacNAc-containing glycans that may be assigned into low-affinity motifs with less stringent parameters.

The exact motifs and family tree structure produced for RCA-I are but one possible model; many might produce similarly accurate descriptions of glycan binding. The algorithm depends solely on the distribution of glycans in the data and in no way guarantees that primary motifs will have visually different structures (as evidenced by the similarities between primary motifs A and D) although this is generally found to be the case as the binding properties tend to be similar for similar structures. For RCA-I, raising the thresholds for splitting produces a less complex tree, reducing the number of fine specificities for A and reducing the number of splits for motifs with less support such as the fine specificities of B. Changes to the guide glycan selection through the use of alternate random number seeds results in slightly different motif structures with the greatest impact on motifs being produced with less support, although the overall conclusions for the output remain unchanged. Overall, this example shows how MotifFinder can provide insights into the rules that underlie the complexities of lectin binding.

Applications

For many researchers, the primary value of the output of this system will be the descriptions of lectin specificities. The detailed information could have fundamental implications for the biology of particular lectin-glycan interactions. For other researchers, the output will be primarily useful in the application of lectins to probe biological samples. Lectins are widely used as probes for the amount or location of glycans, but the conventional mode of interpreting the data is qualitative and imprecise. A more accurate approach would be to take into account the detailed information available from glycan-array analyses. We previously introduced and validated algorithms that make use of lectin models derived from glycan-array data to interpret lectin binding to a biological sample (*36*, *37*). Data acquired from the incubation of a lectin on an unknown sample is interpreted using the output from the glycan-array analyses to produce a quantitative estimate of the glycans in the sample (Figure 7). There are many benefits of this approach: it provides a higher-accuracy estimate of the glycan motifs in a sample; it provides value for researchers with minimal expertise in glycobiology or minimal knowledge about particular lectins; it enables faster, high-throughput data processing; and it provides the option to integrate information from multiple lectins. This latter benefit can have particular importance for glycan analyses. Any individual lectin provides only a glimpse into the motifs that are present, but multiple lectins in a motif can provide greater detail. Additional information could also be obtained by treating the sample with a glycosidase to expose underlying motifs prior to the incubations of multiple lectins. This approach (known as glycan modification and

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probing) (37) promises to be useful for analyzing samples or glycoproteins that are available in limited quantities and for providing insights that are complementary to those obtained using other methods such as mass spectrometry and chromatography.

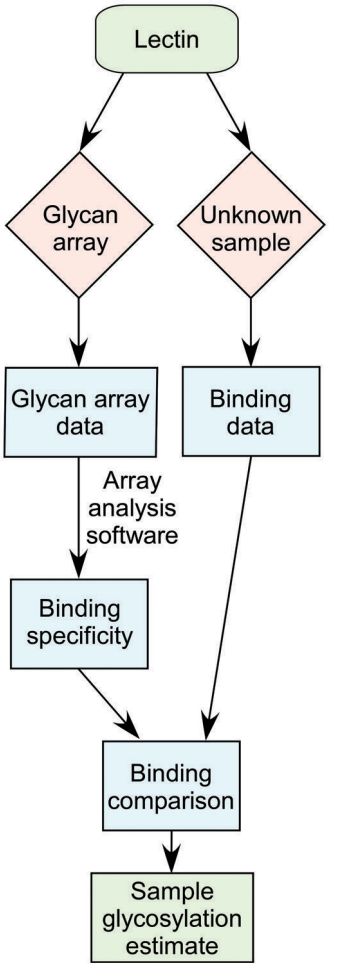


Figure 7. Applying the MotifFinder output to guide the interpretation of lectin binding to unknown samples. In this proposed sequence, the output from glycan-array analyses are compared with the observed binding patterns to an unknown sample.

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